

Darrin Du

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Education

University of Michigan, College of Engineering

Ann Arbor, MI

Bachelor of Science in Engineering (B.S.E.) in Computer Science

May 2028

- GPA: 3.65/4.0 | Honors: Dean's List, Academic Honors, Michigan Achievement Scholarship
- Course Highlights: Data Structures and Algorithms, Operating Systems*, Computer Organization, Computer Vision*

Experience

HAIL Lab - University of Michigan

Ann Arbor, MI

Undergraduate Research Assistant, Software | Python, OpenCV, ROSBAGs, Git/Github

May 2026 - Sept. 2026

- Expand a real-time visual display system in Python and OpenCV, supplying task guidance with ROS sensor data to improve procedural accuracy for clinicians using VIGIL, a **\$26.4M ARPA-H-backed** mobile clinic expanding rural healthcare access.
- **Benchmark** different implementation approaches in 3 core parameters (RealSense positioning, spatial placement, jitter safeguards) to engineer a production design based on measured accuracy and latency tradeoffs rather than assumptions.
- Engineer a dynamic facial overlay and shoulder-anchored tracking system using OpenCV and segmentation algorithms to maintain eye-protection masking during procedures, **incorporating fallback redundancy** for continuous patient safety.
- Partner with hardware and clinical stakeholders to translate ambiguous requirements into concrete technical specs; rapidly prototyped interactive HTML/web concepts using **AI-assisted generation** workflows before hand-off to production engineers.

NASA SUITS - Collaborative Lab for Advancing Work in Space (CLAWS)

Ann Arbor, MI

AR Director | Unity, C#, Mixed Reality Toolkit (MRTK)

Mar. 2026 - Present

- Co-directed a 12-person engineering subteam selected to compete nationally in the NASA SUITS Challenge, overseeing the architecture of an AR astronaut interface and backend.
- Established Git branching standards and mandatory pull-request review across the 12-person subteam, cutting cross-team feature integration time from 14 to 5 days.
- Managed a **2-month onboarding protocol** to teach 4 new members HoloLens 2 deployment workflows and Unity best practices to independently build and see their work within 1 week.

Projects

Clerse

Frontend Developer | Next.js 15, React Flow, TypeScript, Liveblocks, Tailwind CSS

Jan. 2026 - Feb. 2026

- Co-developed a **1st-place** spatial AI canvas platform for a 24-hour Claude hackathon, utilizing **Claude and parallel subagents**.
- Developed an interactive "branching" UI module, allowing users to right-click message threads, fork conversation nodes, and visually track independent contextual branches via animated river edges.
- Executed end-to-end integration testing across 4 UI modules, delivering a flawless live presentation.

GEMINI (CLAWS Sub-System / NASA SUITS)

Mixed Reality Developer | Unity, C#, Mixed Reality Toolkit (MRTK), GitHub

Nov. 2025 - May 2026

- Designed and integrated a cohesive end-to-end Augmented Reality (AR) interface in Unity with MRTK, guiding a **past astronaut** (Anne McClain) and engineers through **5 complex lunar mission protocols**.
- Engineered dynamic UI/UX screen workflows tailored for head-mounted displays (HMDs), **lowering cognitive load** and ensuring stable spatial anchoring under simulated extravehicular activity (EVA) constraints.
- **Bridged the gap between AI and UX** sub-teams, successfully embedding a localized AI agent into the spatial environment of the HoloLens to automate task dispatch and stream **real-time** telemetry.

Synced In

Solo Developer | C#, Unity, 2D Design, Game Development

Sept. 2022 - Dec. 2022

- Architected core game logic, player mechanics, and interactive systems from scratch in C#; published to Unity Play
- Prototyped **20+ puzzle designs** and refined difficulty curves to ship 6 production levels as sole developer, earning **370+ plays**.

Skills

- **Programming Languages:** C++, C#, C, Python, HTML/CSS, TypeScript, R, MATLAB
- **Tools and Technology:** Unity, Mixed Reality Toolkit (MRTK), OpenCV, YOLO, Next.js, React Flow, Version Control
- **Soft Skills:** Cross-Functional Collaboration, Communication & Documentation, Decision-making, Debugging